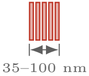
a **three levels; only the middle one is a set**

integrin, one heterodimer.
Not a set — nothing here resolves a molecule.

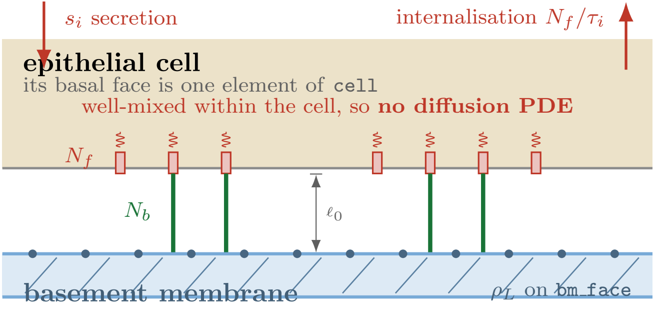


nanocluster, 20–50 integrins.
One of these is one plaque in the discrete model.

adhesion, $\sim 10^3$ integrins.
Clusters at 555 nm centres, spacing set cell-intrinsically.

At 41k sheet nodes that is **25 clusters and $\sim$ 3000 integrins per node**. Above one object per element the discrete description is the approximation and a **density** is exact — which is why the plaque stops being a cluster and becomes a bond density.

b **where each state lives**



$\dot N_b = k_{\text{on}}\, n_f\, \rho_L \left(1 - n_b/n_{\max}\right) - k_{\text{off}}(f)\, N_b, \quad k_{\text{off}}(f) = k^0_{\text{off}}\, e^{f/f_b}$
(a load, not a clock)

$\dot N_f = s_i - N_f/\tau_i$ − binding + unbinding

One `bm` node per dot, **one plaque edge per node**, carrying a bond *density* $n_b = N_b/A$ rather than standing for one cluster. Traction $\propto n_b$, with its reaction returned to the cell face — so momentum still closes.

Conserved: $N_f^{\text{cell}} + \sum_{\text{edges}} N_b$ moves only through s_i and $1/\tau_i$: not by stretch, not by a remesh, not by binding. That is the gate.